

Introduction to Coding Theory
EC5.205 Spring 2026
Mid Sem Exam

5rd April, 2026

Please note:

- There are 2 pages in this question paper.
- No source material is to be used.
- Go through every question carefully before answering. Please write down all the steps in your solutions.

Maximum Marks: 30

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1. A codeword of the code $\{01010, 10101\}$ is transmitted through a binary symmetric channel with crossover probability $p = 0.1$ and minimum distance decoder is applied to the received word. Compute the probability of error. (5 marks)
 2. Show that when one adds an overall parity bit to each codeword of an (n, M, d) code over $\{0, 1\}$ where d is odd, an $(n + 1, M, d + 1)$ code is obtained. Hence, if we define the *extended Hamming code* to be an extension of the Hamming code seen in class, with each codeword appended with a parity bit, what is the minimum distance of the new code? Please give complete explanation for your answer. (6 marks)
 3. Let $\mathcal{C}$ be an (n, k) code over $\mathbb{F}_2$. Let $\mathcal{C}_e$ be the set of even Hamming weight (including zero weight) codewords in $\mathcal{C}$. Then prove that either (i) $\mathcal{C} = \mathcal{C}_e$ or (ii) $\mathcal{C}_e$ is an $(n, k - 1)$ subcode of $\mathcal{C}$. (Note: Subcode of a code is defined as a subspace of the original code.) (5 marks)
 4. Let $\mathcal{C}$ be an $[n, k, d]$ linear code. For $i \in [n]$, let us denote by $\mathcal{C}_i$, the code

$$\mathcal{C}_i = \{(c_1, \dots, c_{i-1}, c_{i+1}, \dots, c_n) : (c_1, \dots, c_n) \in \mathcal{C}\}.$$

Show that $\mathcal{C}_i$ is an $[n - 1, k_i, d_i]$ linear code, where $k_i \geq k - 1$ and $d_i \geq d - 1$.

(5 marks)

5. Show the following: If a (n, k) binary linear code contains all one codeword $\mathbf{1}$, then $A_i = A_{n-i}$, i.e., the number of codewords of weight i equals the number of codewords of weight $n - i$, $\forall i = 0, 1, \dots, n$. (5 marks)
6. Show that a binary linear code with parity-check matrix of the form $H = [A|I]$ is self-dual if, and only if, A is square with $AA^T = I$. (4 marks)